

Asymptotics and inversion for unlabeled interval graphs

Leading equivalent and second order corrections

Report 133 A005975 and A005976

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Abstract

Let I_n and C_n count all and connected unlabeled interval graphs on n vertices, and let F_n be the Fishburn number counting unlabeled interval orders. We prove

$$\begin{aligned} C_n &= \tfrac{1}{2}F_n - F_{n-1} - (\log n + b_0)F_{n-2} + o(F_{n-2}), \\ I_n &= \tfrac{1}{2}F_n - \tfrac{1}{2}F_{n-1} - (\log n + b_0)F_{n-2} + o(F_{n-2}). \end{aligned}$$

Here $b_0 = \log(6/\pi^2) + \gamma + 1$, with Euler’s constant γ . The proof identifies the graph fibers responsible for the logarithm: a three-element order of shape V , or its dual, substituted at a simplicial singleton of an otherwise generic interval order. Their rooted positions number $(2 \log n + O(1))F_n$. Modular decomposition makes their graph fibers exact and bounds all other losses. We give the factorial normalization, a monotone threshold inversion with the rounding kept inside rigorous ceiling brackets, and finite checks from freshly generated Fishburn matrices. A second canonical fiber family and sharper ordinal estimates determine the displayed constant. No all-orders expansion is asserted.

[write] **Status and trust boundary, added 5 October 2026, batch 106.** This is bundle Report 133 of an external AI-assisted research session, filed in ProveIt as the single-source research report `a005975-interval-graphs` (provenance, the sources as the write read them, relation to the repository, reading conventions and the collected non-claims in Section 1.1). It is unrefereed and not formalized; its author line reads “Report 133” and its PDF author field is empty. *What rests on what.* Every asymptotic statement is proved by hand from external inputs stated in Section 2: the Fishburn expansion and two statistics of Hwang and Jin [1], and Gallai’s orientation theorems in the presentation of [2]; these are not re-proved. No asymptotic statement rests on a computation; the finite checks of Section 11 test the constructions at small sizes. *Priority.* The leading equivalent $I_n \sim C_n \sim F_n/2$ is not claimed as new: Hanlon [4], read by the write, lists the asymptotic number of interval graphs as an open problem in 1982, and none of the later sources read states it (Remark 12.2); that search was not exhaustive. The write adds Remark 10.3 (strict monotonicity, and the inverse in the terms of the transseries volume), Remark 12.1 (what Bukh and Jeffs state, and the $O(\log n)$ term of their estimate made explicit), Remark 12.2 (the literature from Hanlon to Bukh–Jeffs) and Section 12.1, the open questions, with the shifted constant b_0 compared with exact counts through $n = 30$.

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1 Main results and scope

All graphs and orders in this report are finite and counted up to isomorphism. Write I_n for the number of interval graphs on n vertices (OEIS A005975), C_n for the connected count (A005976), and F_n for the number of interval orders (A022493). An interval graph is an intersection graph of intervals on a line. An interval order has an interval representation in which $x < y$ precisely when the interval of x lies strictly to the left of that of y . Its incomparability graph is an interval graph; every interval graph arises this way. Put $F_0 = I_0 = 1$ and $C_0 = 0$.

Theorem 1.1. *As $n \rightarrow \infty$,*

$$C_n = \frac{F_n}{2} - F_{n-1} - F_{n-2} \log n + O(F_{n-2}), \quad (1)$$

$$I_n = \frac{F_n}{2} - \frac{F_{n-1}}{2} - F_{n-2} \log n + O(F_{n-2}). \quad (2)$$

Consequently, with $q = 6/\pi^2$,

$$\frac{2C_n}{F_n} = 1 - \frac{\pi^2}{3n} - \frac{\pi^4 \log n}{18n^2} + O(n^{-2}), \quad (3)$$

$$\frac{2I_n}{F_n} = 1 - \frac{\pi^2}{6n} - \frac{\pi^4 \log n}{18n^2} + O(n^{-2}). \quad (4)$$

In particular both sequences are asymptotic to $F_n/2$. The negative logarithmic term in (3)–(4) is part of the theorem, not a fit to a short sequence. Section 9 sharpens the shifted formulas to a little- o remainder and identifies their nonlogarithmic constant.

For an equivalent in conventional factorial form, define

$$K = \frac{6\sqrt{3}}{\pi^{5/2}} e^{\pi^2/12}, \quad a_F = \frac{3}{8} - \frac{17\pi^2}{144} + \frac{\pi^4}{432}, \quad \beta = \frac{\pi^4}{18}. \quad (5)$$

The constant K is approximately 1.3521662450.

Corollary 1.2. *For $a_n = I_n$ take $\alpha = \alpha_I = a_F - \pi^2/6$; for $a_n = C_n$ take $\alpha = \alpha_C = a_F - \pi^2/3$. Then*

$$a_n = Kn! \sqrt{n} q^n \left(1 + \frac{\alpha}{n} - \beta \frac{\log n}{n^2} + O(n^{-2}) \right). \quad (6)$$

Equivalently, both leading terms equal

$$\frac{6\sqrt{6}}{\pi^2} e^{\pi^2/12} n^{n+1} \left(\frac{6}{e\pi^2} \right)^n.$$

The report proves the structural transfer from interval orders to interval graphs, including the constant refinement in Theorem 9.1. The analytic Fishburn estimates and Gallai's orientation theorem are stated precisely as external inputs in Section 2. No historical-priority claim is made. All asymptotic assertions allow a finite initial exceptional range; specifically, the generic-quotient argument uses size $m \geq 5$ and its dominant inflated graphs have size $n = m + 2 \geq 7$.

1.1 Provenance, sources and reading conventions

[write] **Added 5 October 2026, batch 106.** *Provenance.* This report is Report 133 of the session bundle of Reports 1–243 that ProveIt’s docs/incoming intake processed as batch 106: the archive A005975_A005976_Interval_Graph_Asymptotics_and_Inversion_Source.zip (492,968 bytes, wrapper directory report133/, 11 files), arrival commit 60f54ea06, placed in 47fc7a069. The text is the delivered report133.tex (694 lines), shipped as article.tex. The delivered 20-page PDF and the checksum manifest MANIFEST.sha256 are not shipped (the manifest was verified at placement); both are retrievable from the arrival commit. The package names no ProveIt commit, path or report, so the report has no pin; it carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 80 delivered labels with `ivg:` and updated every reference; it added this subsection, the notes marked [write], Remarks 10.3, 12.1 and 12.2, Section 12.1 and three bibliography entries, and typeset the summation range of (40) with `\substack` instead of `\atop` (the same display; this removes an `amsmath` warning). No statement, proof or number of the source was changed: the added remarks are the last numbered statements of their sections and the added displays are unnumbered, so every theorem and equation keeps its delivered number.

The sources, as the write read them (5 October 2026).

- Hwang and Jin [1], the 54-page author PDF that the source cites: equation (1.1) on p. 2, which prints Zagier’s expansion with the first coefficient $c_1 = 11/24 - 17\pi^2/144 + \pi^4/432$ and $O(n^{-2})$ remainder; Theorem 25 on pp. 40–41, whose part (ii) and its proof give the diagonal size and its probability generating function; Corollary 29 on p. 46 (self-dual counts). Located at the cited places; not re-derived. The arXiv version 2 (13 October 2020, 44 pages) numbers these Theorem 22 and Corollary 26 and prints (1.1) with remainder $O(n^{-1})$ only; the journal version is J. Combin. Theory Ser. A 180 (2021) 105413.
- Klavík, Kratochvíl, Krawczyk and Walczak [2], arXiv version 1: Section 4, “Modular decomposition”, Theorems 3–6, stated there as Gallai’s. Located; not re-derived.
- Bukh and Jeffs [3], version 2 (28 June 2023): Theorem 1 (labelled graphs) and the unlabelled remark in Section 5, p. 11; see Remark 12.1.
- Hanlon [4], read in the AMS scan (44 pages), Acan [6], read in full, and Yang and Pippenger [7], abstract only; see Remark 12.2.
- OEIS A005975 (24 terms in the data field), A005976 (25 terms) and A022493 [5], fetched by the intake on 5 October 2026.

What was checked at intake. The batch-106 dossier read the manuscript in full and checked Lemmas 5.1–5.4, Proposition 6.1, the case split of Section 7 and the Newton bound; it found no gap. Against the OEIS it checked that the Euler transform (30) of C gives I through $n = 24$ and that $C_{n+1} \geq I_n \geq C_n$ through $n = 24$, and the delivered checker passed (normal and optimized outputs byte-identical). The write rechecked the arithmetic of the constant in Theorem 9.1 (the second-order ordinal term (42) and the combination $a_0 + \frac{1}{2} + \frac{1}{2}$), reran the checker from copies of the shipped programs (note at the end of Section 11), extended the numerical comparison to $n = 30$ with Hanlon’s tables, and computed the diagonal totals T_m exactly through $m = 200$ (Question 1).

Relation to the repository. Before batch 106 no file of the repository mentioned A005975 or A005976 (A022493 occurs in other reports), and the source makes no claim about the repository, so nothing needed correcting. The report meets the following texts.

- **a336070-weak-ascents** (same collection), a *neighbour*: it uses the same equation (1.1) of [1] as an external input (its `wasc:gr:eq:fishburn`), and the case $d = 0$ of its difference-ascent family is the Fishburn sequence A022493. No shared theorem.
- **a156808-circle-graphs** (bundle Reports 150 and 152; unlabelled circle graphs A156808, A156809) and **a123448-permutation-graphs** (bundle Report 153; unlabelled permutation graphs A123448),

siblings of the same batch. The three share a pipeline, not a theorem: count a representation model (here interval orders, through Fishburn matrices; there indexed chord matchings and permutations); show that almost every graph has a recoverable prime core with bounded decorations whose representation fibre is a full orbit of the symmetry group of the representation (here duality, of order 2, whence $F_n/2$; there the dihedral group of order $4n$ and the Klein four-group); make the symmetric cores negligible; invert. The manuscripts do not cite one another, use different decomposition theories and inputs, and give the letters S_n , F , T , Q , K , β different meanings (table below).

- The transseries volume
`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`
`(transseries_and_inversion.tex)`: Remark 10.3 says which parts of Theorem 10.1 are instances of its statements.

No Lean or Rocq file of the repository treats interval graphs, interval orders in the order-theoretic sense, or Fishburn numbers; no statement of this report is formalized, and its place in the collection confers no formal status.

[write] What is not claimed, collected from the source. No all-orders expansion. The inverse theorem asserts the existence of constants and an onset, supplies a certified value for neither, and is not a finite algorithm for exact thresholds; $N_a(T) = \lceil y \rceil + O(\cdot)$ is not asserted (Remark after Theorem 10.1). No historical-priority claim: the limited source comparison “is not an exhaustive novelty search”; the source did not inspect Hanlon’s article (the write did, Remark 12.2). The nonlogarithmic coefficient in the conventional normalization (6) is not evaluated. The external inputs are not newly proved; page and theorem numbers refer to the author versions. Series identities are formal; no positive radius of convergence is asserted. The finite computations verify exact combinatorial constructions at small sizes, not the asymptotic estimates, an error constant or a rounding certificate; the archive seal is not a digital signature. The write adds: its numerical observations (Section 12.1) are exact counts, ratios of them or floating-point fits, and certify no limit.

Reading conventions (letters the manuscript reuses, and letters that its sibling reports use otherwise, with the tempting false readings; no symbol was renamed).

Symbol	Meaning here (where)	Not to be read as
I_n, C_n	all and connected unlabelled interval graphs, A005975 and A005976; $I_0 = 1, C_0 = 0$	labelled counts (the $f_1(n)$ of [3]); Hanlon's i_n, c_n are the same numbers
F_n	Fishburn numbers A022493: interval orders, Fishburn matrices	$F(x) = \sum n!x^n$ in a123448
S_n, S_n^{conn}	self-dual interval-order types (with connected incomparability graph)	symmetric matchings (a156808); simple permutations $S(z)$ (a123448)
O_n, A_n	orders with connected incomparability graph; $A_n = F_n - O_n$ (Section 7)	O, big-O; the factor $A_m(v)$ in the proof of Lemma 4.2
T_m, R_m	total diagonal weight; total diagonal occupancy (rooted types)	T , the level in $N_a(T)$ (Section 10)
$\mathcal{E}_m, \mathcal{D}_n, \mathcal{Q}_m, \mathcal{J}_n$	eligible rooted types; the dominant graph family; generic non-self-dual types; the serial family	Q_n in a156808
B_m^{all}, B, B'	nongeneric orders; the bracket constants of Theorem 10.1	each other
K, q, a_F, β	the constants (5), $q = 6/\pi^2, \beta = \pi^4/18$	the unknown second prime coefficient β of a156808 ; its truncation K
a_0, b_0, b	$\log q + \gamma; \log q + \gamma + 1; b = \log(K\sqrt{2\pi})$ in the proof of Theorem 10.1	b as b_0
c_1	coefficient in (1.1) of [1], normalization $n^{n+1}(q/e)^n$; $a_F = c_1 - 1/12$	a_F
L, L_j	$\log T$ (Section 10); the exceptional-core count (26)	each other
u, w	number of removed universal vertices (Sections 7, 9); the universal vertex and the leaf of the serial family	each other
M	a module (Section 9); the error constant in the proof of Theorem 10.1	each other
h, V, Λ	the step $(\log m)^{-2}$ or $(\log m)^{-3}$; the three-element orders of Section 6	h_n in a156808

2 Definitions and established inputs

A poset subset M is *autonomous* if every exterior element is below all of M , above all of M , or incomparable to all of M . A graph subset is a *module* if every exterior vertex is adjacent to all or none of it. Two sets overlap if their intersection and both set differences are nonempty. A graph module is *strong* if it overlaps no graph module. At the root of the strong-module decomposition, the maximal proper strong modules partition the vertices; we call them the *root children* or *bags*. The root quotient records adjacency between bags. A graph is *prime* if it has no proper nonsingleton module.

For a poset P write $H(P) = \text{Comp}(P)$ and $G(P) = \text{Inc}(P)$. An interval order is equivalently a poset with no induced $2 + 2$, namely two unrelated two-element chains. Consequently $H(P)$ has no induced $2K_2$: orienting two disjoint edges transitively would produce a $2 + 2$. Duality reverses all order relations; write P^* for the dual and S_n for the number of self-dual interval-order types of size n .

A *Fishburn matrix* is an upper triangular matrix of nonnegative integers with no zero row or column. Its size is the sum of its entries. The standard bijection with interval orders assigns A_{ij} indistinguishable elements to the interval $[i, j]$, with $x < y$ when the right endpoint of x is smaller than the left endpoint of y . Thus there are F_n such matrices of size n . The no-zero-row and no-zero-column conditions recover the canonical chains of strict down-set and up-set levels; this makes the correspondence one-to-one on isomorphism types.

We use the following established results.

1. **Fishburn asymptotics.** Hwang and Jin [1], equation (1.1), printed page 2 of the cited author version, state the expansion attributed there to Zagier. In the present normalization it is

$$F_n = 2Kn! \sqrt{n} q^n \left(1 + \frac{a_F}{n} + O(n^{-2}) \right). \quad (7)$$

Their coefficient in the normalization $n^{n+1}(q/e)^n$ is $c_1 = 11/24 - 17\pi^2/144 + \pi^4/432$; Stirling's formula subtracts $1/12$, giving $a_F = c_1 - 1/12$.

2. **Diagonal statistic.** If Y_m is the total diagonal weight of a uniformly selected size- m Fishburn matrix, Theorem 25(ii) and the displayed probability-generating-function estimate in its proof, printed pages 40–41 of [1], give

$$\mathbb{E}v^{Y_m} = \Gamma(v)^{-2}(qm)^{2(v-1)} \left(1 + O(m^{-1} \log m) \right) \quad (8)$$

uniformly for real v in a fixed neighborhood of 1. Real uniformity suffices for this report.

3. **Self-duality.** Corollary 29 on printed page 46 of [1], counting by ordinary size rather than reduced size, implies

$$\log S_n = \frac{1}{2}n \log n + O(n), \quad S_n = o(F_n n^{-A}) \quad \text{for every fixed } A > 0. \quad (9)$$

The second statement follows immediately from (7) and Stirling's formula.

4. **Gallai orientation decomposition.** We use the statements presented as Theorems 3–6 in Section 4, printed pages 5–7, of Klavík, Kratochvíl, Krawczyk and Walczak [2]. Transitive orientations are uniform between adjacent disjoint strong modules, are specified by independent transitive orientations of strong-module quotients, and have root quotient empty, complete, or prime. A prime comparability quotient has exactly two transitive orientations, mutually reversed. In particular every strong module of a comparability graph is autonomous in every transitive orientation, by comparing it with exterior singleton modules.

Page and theorem numbers refer to the linked author versions; published versions may differ. These are analytic and structural inputs, not claims newly proved here. All subsequent graph-fiber and error estimates are proved below.

3 Factorial estimates

From (7), after enlarging constants at finitely many indices, there are positive c, C with

$$cq^j j! \sqrt{j} \leq F_j \leq Cq^j j! \sqrt{j} \quad (j \geq 1). \quad (10)$$

Lemma 3.1. *For fixed positive integers r, s and fixed real a, b ,*

$$\sum_{k=r}^{N-s} k!(N-k)!k^a(N-k)^b = O\left(N! \left[N^{b-r} + N^{a-s}\right]\right). \quad (11)$$

Proof. Split the sum at $N/4$ and $3N/4$. In the first range the consecutive-term ratio is

$$\frac{k+1}{N-k} \left(\frac{k+1}{k}\right)^a \left(\frac{N-k-1}{N-k}\right)^b.$$

After a fixed initial segment, and for large N , it is bounded by a constant strictly below 1. The fixed initial terms have the scale of the first endpoint, $N!N^{b-r}$; a geometric tail has the same bound. The final range is symmetric and gives $N!N^{a-s}$. In the middle, $k!(N-k)! = N!/ \binom{N}{k}$ is exponentially small relative to $N!$, uniformly in the range. It dominates every fixed polynomial weight and the number of summands. Empty ranges and finitely many small N are absorbed into the constants. $\square$

We will use the following consequences of (10) and Lemma 3.1:

$$\sum_{k=1}^{n-1} F_k F_{n-k} = O(F_{n-1}), \quad \sum_{k=2}^{n-2} F_k F_{n-k} = O(F_{n-2}), \quad (12)$$

$$\sum_{k=3}^{n-2} F_k F_{n-k+1} = O(F_{n-2}), \quad \sum_{k=4}^{n-3} F_k F_{n-k+1} = O(F_{n-3}). \quad (13)$$

For the second line take $N = n + 1$ and exponents $1/2, 1/2$; its endpoint pairs are $(3, 3)$ and $(4, 4)$. The adjacent ratio from (7) also gives

$$\sum_{j=0}^N F_j = O(F_N), \quad \sum_{j=1}^N j F_j = O(N F_N). \quad (14)$$

For example, beyond a fixed index successive terms grow by at least a factor 2, so the reverse sum is geometrically bounded. Fixed powers of $\log(j+1)$ can be inserted in these prefix sums at the cost of the corresponding factor at N . We also record

$$\frac{F_{n-1}}{F_n} = \frac{1}{qn} (1 + O(n^{-1})), \quad \frac{F_{n-2}}{F_n} = \frac{1}{q^2 n^2} (1 + O(n^{-1})). \quad (15)$$

4 Simplicial roots and diagonal occupancy

A graph vertex is *simplicial* when its open neighborhood is a clique. Let T_m be the sum of the numbers of simplicial vertices over all size- m interval orders. Let R_m count isomorphism types of size- m interval orders with one simplicial vertex distinguished. These are different counts: the latter counts vertex orbits under order automorphisms.

Lemma 4.1. *A vertex of a Fishburn order is simplicial in its incomparability graph exactly when it lies in a diagonal cell. The automorphism orbits of the order are exactly its occupied matrix cells. Thus T_m is total diagonal weight over the matrices and R_m is total diagonal occupancy. With $T_0 = 0$,*

$$T_m - R_m = T_{m-1} \quad (m \geq 1). \quad (16)$$

Proof. Every interval meeting $[i, i]$ contains i , so a diagonal vertex is simplicial. If a vertex has interval $[i, j]$ with $i < j$, the nonzero column i supplies an element in a cell (a, i) and the nonzero row j supplies one in (j, b) . They both meet $[i, j]$ but are disjoint from one another. Hence that vertex is not simplicial.

The distinct strict down-sets of an interval order form a chain by inclusion; so do its distinct strict up-sets. Every automorphism preserves each level of both chains. The two level indices identify a matrix cell, so automorphisms preserve cells. Conversely, arbitrary permutations of elements within a cell preserve all relations. These are precisely the orbits.

A diagonal cell of weight $r \geq 2$ contributes $r - 1$ to $T_m - R_m$. Decrease its entry by exactly one and keep that cell marked with one of $r - 1$ auxiliary indices. Its support does not change, so the result is a size- $(m - 1)$ Fishburn matrix with a positive diagonal cell of weight $r - 1$ and one of its $r - 1$ weight units distinguished. Conversely, increase the entry of any such marked cell by one. The auxiliary index is retained. This is a weighted bijection: the target count is exactly T_{m-1} . Cells of weight one contribute zero to the difference. For $m = 1$ the same identity uses the empty size-zero matrix and $T_0 = 0$. $\square$

The auxiliary weight unit in the last proof is a counting device, not a claim that its choices give different rooted isomorphism types.

Lemma 4.2. *As $m \rightarrow \infty$,*

$$T_m = (2 \log m + O(1))F_m, \quad R_m = (2 \log m + O(1))F_m. \quad (17)$$

In fact $T_m - R_m = O(F_{m-1} \log m) = o(F_m)$.

Proof. We extract the mean from (8) without differentiating its unspecified remainder. Set $h = (\log m)^{-2}$. For integer $Y \geq 0$, Bernoulli's inequalities give

$$\frac{1 - \mathbb{E}(1 - h)^Y}{h} \leq \mathbb{E}Y \leq \frac{\mathbb{E}(1 + h)^Y - 1}{h}.$$

For $A_m(v) = \Gamma(v)^{-2}(qm)^{2(v-1)}$, Taylor expansion at 1 has linear coefficient $2 \log(qm) + 2\gamma$ and quadratic remainder $O(h^2(\log m)^2)$ uniformly at $1 \pm h$. Here γ is Euler's constant. The multiplicative error in (8), divided by h , contributes $O((\log m)^3/m) = o(1)$; $A_m(1 \pm h)$ stays bounded. Both bounds for $\mathbb{E}Y_m$ therefore equal $2 \log m + O(1)$. Multiplication by F_m proves the assertion for T_m . Lemma 4.1 and (15) then give the other statements. $\square$

5 Substitution and generic quotients

To substitute a poset R for a vertex v of a poset Q , replace v by R and give every element of R exactly the old relation of v to each exterior element.

Lemma 5.1. *Let Q and R be interval orders. If R contains a comparable pair, substitution at v produces an interval order if and only if v is simplicial in $G(Q)$. If R is an antichain, substitution always produces an interval order.*

Proof. If two exterior elements $y < z$ are both incomparable to v , they and a comparable pair in R induce a $2 + 2$ after substitution. This proves necessity.

For sufficiency, suppose a substituted order contains an induced $2 + 2$ on a set X . With zero, one, or four vertices of X in R , an induced $2 + 2$ would already occur in Q or R . With three vertices in R , the exterior vertex is comparable to all three or to none; in a $2 + 2$ every vertex has exactly one comparability, a contradiction. With two vertices in R , any exterior-to-interior comparability forces an exterior vertex to be comparable to both interior vertices, also impossible. Thus the two chains must be an internal pair and an exterior pair. The latter pair is comparable and both its vertices are incomparable to v , contradicting simpliciality. If R is an antichain, the final internal comparable pair is impossible regardless of v . $\square$

Lemma 5.2. *Substitute nonempty graphs B_i at the vertices of a prime graph J of size at least four. The root children of the resulting graph are exactly the bags B_i .*

Proof. Every bag is a module. If a graph module N meets at least two bags, the set S of vertices of J corresponding to met bags is a module of J : any unmet bag has uniform adjacency to all met bags, by the module property of N . Primality implies $S = V(J)$. If N omits a vertex of a bag B_i , that omitted vertex must have uniform adjacency to N , so i is adjacent either to all other vertices of J or to none. A prime graph of size at least four has no universal or isolated vertex, since the remaining vertices would form a proper nonsingleton module. Thus N omits no vertex and is the whole graph. Every proper module consequently lies within one bag. This proves that the bags are strong and maximal proper strong modules. $\square$

Call a size- m interval order *generic* in this report if it has no autonomous subset of any kind of size between 3 and $m - 1$. This definition includes antichain modules; it is stronger than merely excluding large modules containing comparable pairs.

Lemma 5.3. *The number B_m^{all} of nongeneric interval orders is $O(F_m/m)$. For $m \geq 5$, a generic interval order has a prime comparability root quotient, and all root children have size at most two. Its comparability and incomparability graphs are both connected; its incomparability graph has exactly one dual orbit of interval-order types above it.*

Proof. Choose a forbidden autonomous subset of size k , contract it to a distinguished representative, and retain its internal interval order. The quotient is an induced interval order of size $m - k + 1$, and its rooted type together with the internal type reconstructs the original order uniquely. There are at most $(m - k + 1)F_{m-k+1}$ rooted quotients. Nonunique choices only overcount, so

$$B_m^{\text{all}} \leq \sum_{k=3}^{m-1} (m - k + 1)F_{m-k+1}F_k = O(F_m/m). \quad (18)$$

For the estimate, use Lemma 3.1 with $N = m + 1$, exponents $1/2, 3/2$, and endpoints $r = 3, s = 2$. The result is $O(q^{m+1}(m+1)!(m+1)^{-3/2})$. At the large-module endpoint the rooted quotient has size two, not size m .

Every proper strong graph module is autonomous, so a generic order has root children of size at most two. If the root quotient is empty, take a union of children with total size 3 or 4; it is a proper autonomous subset when $m \geq 5$. Such a union exists by accumulating children until reaching size at least 3. If the quotient is complete, its orientation linearly orders the children; a consecutive prefix with total size 3 or 4 is again proper and autonomous. Both cases contradict genericity, so the quotient is prime. It and its complement are connected, and substitution of nonempty bags preserves these two connections.

Gallai's theorem gives two reversed orientations at the prime root. Within a two-element bag there is a unique order type: an antichain, or a chain whose two orientations are interchanged by swapping its vertices. Those swaps extend to graph automorphisms since the bag is a module. Therefore every interval-order type above this graph is isomorphic to the given order or its dual. Both occur, so there is exactly one dual orbit. $\square$

As an immediate leading-order check, all F_m orders have $(F_m + S_m)/2$ dual orbits, and their graph map is onto. The generic orders map into connected graphs with at most two order types per fiber. Thus

$$\frac{F_m - B_m^{\text{all}}}{2} \leq C_m \leq I_m \leq \frac{F_m + S_m}{2},$$

which already proves $I_m, C_m = (F_m/2)(1 + O(m^{-1}))$.

For $m \geq 5$ define $\mathcal{E}_m$ to be the family of simplicially rooted interval-order types (Q, v) satisfying all three conditions:

1. Q is generic;
2. v belongs to no autonomous two-element subset;
3. Q is not self-dual.

The second condition makes v a singleton root child. Duality acts freely on $\mathcal{E}_m$.

Lemma 5.4. *As $m \rightarrow \infty$,*

$$|\mathcal{E}_m| = (2 \log m + O(1))F_m, \quad R_m - |\mathcal{E}_m| = O(F_m). \quad (19)$$

Proof. The nongeneric orders exclude at most $mB_m^{\text{all}} = O(F_m)$ rooted types. If a marked vertex belongs to an autonomous pair, contract a chosen such pair. The quotient has size $m - 1$ with its contraction vertex distinguished; its internal two-element order has the original root marked. There are three internal rooted types: the antichain, a chain marked below, and a chain marked above. This bounds the excluded types by $3(m - 1)F_{m-1} = O(F_m)$. Choices of pairs merely overcount. Finally self-dual orders exclude at most $mS_m = o(F_m)$ rooted types. Apply Lemma 4.2. No assumption of trivial automorphism groups is needed. $\square$

6 The exact fibers producing the logarithm

Let V be the three-element order with one lower element below two incomparable upper elements; let $\Lambda = V^*$ be its dual. Their comparability graph is $K_{1,2}$. They are nonisomorphic and are the only two interval-order types with that comparability graph.

For $(Q, v) \in \mathcal{E}_m$, substitute either V or Λ at v . Lemma 5.1 gives interval orders on $n = m + 2$ vertices, both with the same incomparability graph. Let $\mathcal{D}_n$ be the family of these graph types. This construction and the conclusions below use $m \geq 5$, hence $n \geq 7$.

Proposition 6.1. *Every graph in $\mathcal{D}_n$ is connected and has exactly four interval-order types above it, comprising exactly two dual orbits and no self-dual order. The map $\mathcal{E}_{n-2} \rightarrow \mathcal{D}_n$ is exactly two-to-one. Consequently*

$$|\mathcal{D}_n| = \frac{1}{2}|\mathcal{E}_{n-2}| = F_{n-2} \log n + O(F_{n-2}). \quad (20)$$

Proof. The prime comparability root quotient of Q survives the substitution. By Lemma 5.2, its root children are the old bags except that the singleton $\{v\}$ becomes a three-vertex $K_{1,2}$ bag. Every other bag has size at most two. The new bag is therefore the unique three-element root child, identified intrinsically from the graph. Contracting it recovers the rooted graph $(G(Q), v)$. Both the comparability graph and its complement remain connected.

Gallai's orientation decomposition permits just two root orientations, and all unchanged bags have unique order type. The exceptional bag permits V or Λ . Thus there are at most four order types, represented by

$$Q[V], \quad Q[\Lambda], \quad Q^*[V], \quad Q^*[\Lambda].$$

All four exist by Lemma 5.1. An isomorphism must preserve the unique three-element strong root child. On contracting it, a type with quotient Q cannot become one with quotient Q^* because Q is not self-dual. With quotient fixed, restricting an isomorphism to the exceptional child cannot identify V with Λ . Thus all four types are distinct. Duality pairs $Q[V]$ with $Q^*[\Lambda]$ and $Q[\Lambda]$ with $Q^*[V]$, so none is self-dual.

The two rooted types (Q, v) and (Q^*, v) give the same graph. Conversely, the unique exceptional child canonically recovers the rooted quotient graph. The proof of Lemma 5.3 works in this rooted setting: the only internal swaps required are in two-element bags, and they fix the singleton root. Its interval-order types are exactly (Q, v) and (Q^*, v) , distinct by non-self-duality. This proves the two-to-one statement even in the presence of arbitrary graph automorphisms. Apply (19) and $\log(n-2) = \log n + O(n^{-1})$. $\square$

For a graph with t dual orbits of orders above it, identifying the orbits into one graph loses $t-1$ counts. Proposition 6.1 says that each graph in $\mathcal{D}_n$ contributes a loss of exactly one.

7 All other fiber losses

Let O_n count interval orders with connected incomparability graph, and $A_n = F_n - O_n$. Let S_n^{conn} count the self-dual types among them. The number of their dual orbits is $(O_n + S_n^{\text{conn}})/2$. Define the total connected graph-fiber loss by

$$\Delta_n = \frac{O_n + S_n^{\text{conn}}}{2} - C_n \geq 0. \quad (21)$$

We will prove

$$\Delta_n = F_{n-2} \log n + O(F_{n-2}). \quad (22)$$

7.1 Ordinal decomposition

Every nonempty poset has a unique ordinal-sum decomposition into its incomparability components: any two such components have a uniform comparability direction, and these directions linearly order the components. To see uniformity, an incomparability edge within one component forces its two endpoints to have the same relation to an element of another component; otherwise transitivity would make the endpoints comparable. Propagate along paths in each component. Transitivity then orders the components. Each component of an interval order remains an interval order. Consequently, as formal series,

$$\sum_{n \geq 0} F_n z^n = \frac{1}{1 - \sum_{n \geq 1} O_n z^n}, \quad A_n = \sum_{k=1}^{n-1} O_k F_{n-k}. \quad (23)$$

By (12), $A_n = O(F_{n-1})$. Separating the endpoints $k = 1, n - 1$ for large n gives

$$A_n = F_{n-1} + O_{n-1} + O(F_{n-2}) = 2F_{n-1} + O(F_{n-2}), \quad O_n = F_n - 2F_{n-1} + O(F_{n-2}). \quad (24)$$

Here ordinal indecomposability means connectedness of the incomparability graph; it is not a claim of connectedness of the comparability graph.

7.2 Removing universal graph vertices

Consider a connected interval graph G and any order P above it. The comparability graph H has no induced $2K_2$, so at most one of its components has an edge. All remaining components are isolated vertices. If there is no edge, G is complete; its fiber consists of the antichain, with zero loss.

Otherwise remove the $u \geq 0$ isolated vertices of H , exactly the universal vertices of G . The remaining H_0 is connected with an edge. Let P_0 be the core order and G_0 its incomparability graph; G_0 need not be connected. Restoring the removed vertices adds u elements incomparable to every core element. Thus removing and restoring them gives a bijection of entire order fibers, commuting with duality. The integer u is intrinsic to G .

7.3 A disconnected incomparability core

If G_0 is disconnected, then P_0 is ordinal decomposable and $u \geq 1$, since G is connected. Counting every such core, even without requiring connected H_0 , bounds the number of relevant size- n orders by

$$\sum_{j=1}^{n-1} A_j = O(F_{n-2}) \quad (25)$$

using (24) and (14). This also bounds their fiber losses.

7.4 A prime core with a removed universal vertex

Suppose both H_0 and G_0 are connected. Gallai's classification makes the root quotient prime, with at least four children. If every child is an antichain of arbitrary size or a two-element chain, every child has a unique order type and the two reversed root orientations give just one dual orbit. Such a fiber loses nothing.

Otherwise an edge-containing root child has size $k \geq 3$. With core size j , at least three other children imply $k \leq j - 3$. Contraction gives a simplicially rooted interval order, by Lemma 5.1. Rooted isomorphism types are at most $R_{j-k+1} \leq T_{j-k+1}$, so the number of exceptional core orders is at most

$$L_j := \sum_{k=3}^{j-3} T_{j-k+1} F_k = O(F_{j-2} \log(j+1)). \quad (26)$$

For the estimate, bound T_t by $O(F_t \log(t+1))$ and use (13), harmlessly enlarging the upper limit to $j - 2$. When $u \geq 1$, $j \leq n - 1$, and the total bound is

$$\sum_{j=1}^{n-1} L_j = O(F_{n-3} \log(n+1)) = O(F_{n-2}). \quad (27)$$

The last equality follows from (15). This controls any number of removed universal vertices.

7.5 No universal vertex and a large comparable child

Now the prime core has size n . Orders having an edge-containing root child of size at least four number at most

$$\sum_{k=4}^{n-3} T_{n-k+1} F_k = O(F_{n-3} \log(n+1)) = O(F_{n-2}). \quad (28)$$

Use the second estimate of (13). The cutoff $n-3$ follows from the four or more root children and is essential to this bound. Antichain children of arbitrary size are not charged here; their internal order type is unique.

7.6 The remaining three-element children

It remains to consider prime-root graphs whose edge-containing children have size at most three. Among comparability graphs with at most three vertices, only $K_{1,2}$ has more than one interval-order type, namely V and Λ . The other possibilities are an empty graph, one edge and an isolated vertex, and K_3 , with the unique types antichain, chain plus unrelated singleton, and three-element chain, respectively. Thus if no root child is $K_{1,2}$, the graph fiber has one dual orbit.

Otherwise choose a $K_{1,2}$ root child and contract it, obtaining a simplicially rooted interval order (Q, v) of size $m = n - 2$. If $(Q, v) \in \mathcal{E}_m$, the original graph belongs to $\mathcal{D}_n$ and was counted exactly in Proposition 6.1. If the graph is outside $\mathcal{D}_n$, every such contraction lies in the excluded rooted family. For each excluded rooted quotient there are only the two internal types V, Λ . Therefore the remaining order types number at most

$$2(R_m - |\mathcal{E}_m|) = O(F_m) = O(F_{n-2}). \quad (29)$$

This bound handles multiple, nested, and interacting exceptional modules without assuming independence. Any chosen contraction into $\mathcal{E}_m$ would put the graph in $\mathcal{D}_n$; otherwise it is covered by (29). The finitely many sizes below $m = 5$ do not affect the asymptotic estimate.

Proof of (22). For every positive-loss graph outside $\mathcal{D}_n$, all its order types lie in the cases bounded above. If one of its types instead satisfies a stated one-dual-orbit condition, that condition proves the whole graph fiber has zero loss, a contradiction. A fiber with t order types loses at most t , so (25)–(29) bound the total loss outside $\mathcal{D}_n$ by $O(F_{n-2})$. Each graph in $\mathcal{D}_n$ loses exactly one, and (20) completes the proof. $\square$

8 Passing to connected and all graph counts

Proof of Theorem 1.1 and Corollary 1.2. Equations (21), (24), and (22), together with $0 \leq S_n^{\text{conn}} \leq S_n = o(F_{n-2})$, give

$$C_n = \frac{O_n + S_n^{\text{conn}}}{2} - \Delta_n = \frac{F_n}{2} - F_{n-1} - F_{n-2} \log n + O(F_{n-2}).$$

For all graphs, disconnected graphs with a component of size $n-1$ are exactly a connected graph of that size plus an isolated vertex. Every other disconnected graph can have its components partitioned into two groups of sizes $k, n-k$ with $2 \leq k \leq n-2$: take a component of size at least

two if one exists, or two isolated vertices otherwise. This encodes the graph nonuniquely as a pair of interval graphs. Since $I_j \leq F_j$, for $n \geq 4$ we obtain

$$0 \leq I_n - C_n - C_{n-1} \leq \sum_{k=2}^{n-2} I_k I_{n-k} \leq \sum_{k=2}^{n-2} F_k F_{n-k} = O(F_{n-2}).$$

The connected formula at $n-1$ gives $C_{n-1} = F_{n-1}/2 + O(F_{n-2})$, proving (2). For reference the exact unlabeled component identity is

$$\sum_{n \geq 0} I_n z^n = \prod_{k \geq 1} (1 - z^k)^{-C_k}, \quad (30)$$

since a graph is a multiset of connected components. All series identities are formal; no positive radius of convergence is asserted.

Dividing (1)–(2) by $F_n/2$ and applying (15) gives (3)–(4), with common logarithmic coefficient $-2q^{-2} = -\pi^4/18$. Multiplying by (7) gives (6). Products involving $\log n/n^3$ lie within the stated $O(n^{-2})$ remainder. Stirling's leading equivalent gives the alternative normalization. $\square$

Dropping the now identified logarithmic term gives the previously weaker first-correction precision

$$C_n = F_n/2 - F_{n-1} + O(F_{n-2} \log n), \quad I_n = F_n/2 - F_{n-1}/2 + O(F_{n-2} \log n).$$

It is not valid to transfer the entire Fishburn expansion merely by dividing by two: the graph fibers produce their own inverse-power and logarithmic corrections.

9 The constant after the logarithmic term

The earlier error estimates can be sharpened enough to determine the constant in the exact shifted-Fishburn formulas. This section supplies those refinements; no additional external input is needed. Define

$$a_0 = \log q + \gamma, \quad b_0 = \log q + \gamma + 1. \quad (31)$$

Theorem 9.1. *As $n \rightarrow \infty$,*

$$C_n = \frac{F_n}{2} - F_{n-1} - (\log n + b_0)F_{n-2} + o(F_{n-2}), \quad (32)$$

$$I_n = \frac{F_n}{2} - \frac{F_{n-1}}{2} - (\log n + b_0)F_{n-2} + o(F_{n-2}). \quad (33)$$

The exact shifts in F are part of these formulas. Their constant has three contributions: the constant in the rooted diagonal mean, a second graph-fiber family of loss $F_{n-2}/2$, and the next ordinal-decomposition term.

9.1 A sharper rooted estimate

In Lemma 4.2, use $h = (\log m)^{-3}$ instead of $(\log m)^{-2}$. The same Bernoulli bounds and uniform PGF estimate give, after division by h , the errors

$$O(h(\log m)^2) = O((\log m)^{-1}), \quad O\left(\frac{\log m}{mh}\right) = O\left(\frac{(\log m)^4}{m}\right).$$

Both are $o(1)$. Since the linear coefficient of the main PGF factor is $2\log(qm) + 2\gamma$, the result, together with (16), is

$$T_m = (2\log m + 2a_0 + o(1))F_m, \quad R_m = (2\log m + 2a_0 + o(1))F_m. \quad (34)$$

Indeed $T_{m-1} = O(F_{m-1} \log m) = o(F_m)$, so passing from weight to occupancy changes no constant at the F_m scale.

The exclusions defining $\mathcal{E}_m$ admit the stronger estimate

$$R_m - |\mathcal{E}_m| = O\left(\frac{F_m \log(m+1)}{m}\right) = o(F_m). \quad (35)$$

Here is the complete encoding argument. For a nongeneric rooted order (Q, v) choose a forbidden autonomous subset M of size k , and put $\ell = m - k + 1$.

- If $v \in M$, its restriction remains simplicial in the induced internal order. There are at most T_k internal rooted types and ℓF_ℓ quotients with the contraction vertex marked. These data reconstruct (Q, v) , giving at most $\ell F_\ell T_k$ types.
- If $v \notin M$, its image is simplicial in the induced quotient obtained by keeping a representative of M . This quotient has a simplicial mark and a distinct arbitrary contraction mark, so there are at most $\ell R_\ell \leq \ell T_\ell$ quotient types. With the internal order there are at most $\ell T_\ell F_k$ encodings.

Both bounds concern isomorphism types; allowing every module choice overcounts harmlessly. Since $T_j = O(F_j \log(j+1))$, summing over $3 \leq k \leq m-1$ and using (18) gives

$$\sum_{k=3}^{m-1} (\ell F_\ell T_k + \ell T_\ell F_k) = O(\log(m+1)) \sum_{k=3}^{m-1} \ell F_\ell F_k = O(F_m \log(m+1)/m).$$

If the marked simplicial vertex belongs to an autonomous pair, contracting that pair leaves a simplicial contraction vertex: its exterior neighbors were already neighbors of the original marked vertex and formed a clique. There are only three internal rooted pair types, so this exclusion is at most $3R_{m-1} = O(F_m \log(m+1)/m)$. The self-dual exclusion remains at most mS_m , negligible on this scale. These arguments prove (35), and hence

$$|\mathcal{E}_m| = (2\log m + 2a_0 + o(1))F_m. \quad (36)$$

Proposition 6.1 remains exact, so its loss family now satisfies

$$|\mathcal{D}_n| = (\log n + a_0 + o(1))F_{n-2}. \quad (37)$$

These graphs have no universal vertex, since their comparability graphs are connected.

9.2 A second exact loss family

Let $\mathcal{Q}_m$ consist of the generic, non-self-dual interval-order types of size $m \geq 5$. Lemma 5.3 and (9) give

$$|\mathcal{Q}_m| = F_m + O(F_m/m). \quad (38)$$

For $Q \in \mathcal{Q}_m$, take $G(Q)$ disjoint union a singleton w , and add a new vertex u adjacent to every existing vertex. Call the resulting graph family $\mathcal{J}_{m+2}$.

Both special vertices are graph-intrinsic. The vertex u is the unique universal vertex: every vertex in $G(Q)$ misses w , and w misses $G(Q)$. The vertex w is the unique leaf: $G(Q)$ is connected on at least five vertices, so each of its vertices has at least one neighbor there and degree at least two after adding u . Removing u, w therefore recovers $G(Q)$ canonically.

The fiber of $G(Q)$ consists exactly of Q, Q^* , by its prime root and small children and the non-self-duality of Q . After removing u , the two incomparability components are $G(Q)$ and the singleton. Their ordinal order can be chosen in either direction. Thus the full fiber of a graph in $\mathcal{J}_{m+2}$ has exactly the four types

$$Q + 1, \quad 1 + Q, \quad Q^* + 1, \quad 1 + Q^*,$$

with u incomparable to all elements understood. The symbol $+$ denotes ordinal sum. The position of the singleton component and the non-self-duality of Q make these four types distinct. Duality pairs the first with the fourth and the second with the third. There are two dual orbits, no self-dual type, and a loss of exactly one.

Canonical removal of u, w also proves that $\mathcal{Q}_m \rightarrow \mathcal{J}_{m+2}$ is exactly two-to-one. Hence

$$|\mathcal{J}_n| = \frac{1}{2}|\mathcal{Q}_{n-2}| = \frac{1}{2}F_{n-2} + o(F_{n-2}). \quad (39)$$

The families $\mathcal{J}_n$ and $\mathcal{D}_n$ are disjoint because only the former has a universal vertex. These structural assertions use $m \geq 5$, or $n \geq 7$, as before.

9.3 The remaining loss is little o

We refine the four-case exhaustion of Section 7. First note the triple convolution

$$\sum_{\substack{r,s,t \geq 1 \\ r+s+t=j}} F_r F_s F_t = O(F_{j-2}). \quad (40)$$

Apply (12) to the inner sum over $s + t$; the remaining sum is $O(\sum_{r=1}^{j-2} F_r F_{j-r-1}) = O(F_{j-2})$ by the same estimate with total $j - 1$.

For a disconnected incomparability core, write u for the number of removed universal vertices as before. If $u \geq 2$, its possible sizes are at most $n - 2$, so (23) and (14) give only $O(F_{n-3})$ orders. If $u = 1$, the core size is $j = n - 1$ and we distinguish its ordinal components.

- At least three components can be encoded by the first two components and the remaining nonempty suffix. By (40) there are at most $O(F_{j-2}) = O(F_{n-3})$ such orders.
- Exactly two components, both of size at least two, give at most $\sum_{k=2}^{j-2} F_k F_{j-k} = O(F_{n-3})$ orders.
- The only remaining case is a singleton component beside a component Q of size $m = n - 2$. If $Q \in \mathcal{Q}_m$, the graph lies in $\mathcal{J}_n$. Otherwise Q is nongeneric or self-dual, and the two possible ordinal positions give at most $2(B_m^{\text{all}} + S_m) = O(F_m/m) = O(F_{n-3})$ orders.

Thus the entire disconnected-core contribution outside $\mathcal{J}_n$ is $o(F_{n-2})$.

For a prime core with $u \geq 1$, the earlier bound (27) is already $O(F_{n-3} \log(n+1)) = o(F_{n-2})$. For no universal vertex and a comparable child of size at least four, (28) has the same little- o precision. Large antichain bags still have unique internal type and incur no charge.

In the final case, every comparable root child has size at most three. The reasoning of Section 7 shows that any positive-loss graph outside $\mathcal{D}_n$ contracts at a chosen $K_{1,2}$ child into the excluded rooted family. The stronger bound (35) makes the total at most

$$2(R_{n-2} - |\mathcal{E}_{n-2}|) = O(F_{n-2} \log(n+1)/n) = o(F_{n-2}).$$

Multiple or nested choices only overcount. As in the earlier exhaustion, every order in a positive-loss fiber outside the two exact families must lie in these exceptional cases: a nonexceptional order would certify a single dual orbit for the whole graph. Loss is at most the number of order types, so the order-count bounds control total loss. Equations (37) and (39) now yield

$$\Delta_n = (\log n + a_0 + \tfrac{1}{2} + o(1))F_{n-2}. \quad (41)$$

9.4 Ordinal and component corrections at the next shift

Since $O_1 = O_2 = 1$, $F_1 = 1$, and $F_2 = 2$, isolate the four endpoints $k = 1, 2, n-2, n-1$ of (23). For sufficiently large n , the remaining convolution has both factors of size at least three, hence is $O(F_{n-3})$. Therefore

$$A_n = F_{n-1} + F_{n-2} + 2O_{n-2} + O_{n-1} + O(F_{n-3}).$$

Use (24) at $n-1$ and $n-2$ to obtain

$$O_n = F_n - 2F_{n-1} - F_{n-2} + O(F_{n-3}). \quad (42)$$

Together with (21), (41), and negligible S_n^{conn} , this gives

$$C_n = \tfrac{1}{2}F_n - F_{n-1} - (\log n + a_0 + 1 + o(1))F_{n-2},$$

proving (32). The extra 1 combines the $1/2$ loss from the serial family with the $-F_{n-2}/2$ in $O_n/2$.

For all graphs, those with a component of size n , $n-1$, or $n-2$ contribute C_n , C_{n-1} , or $2C_{n-2}$, respectively, for sufficiently large n . The factor two is $I_2 = 2$, for an edge or two isolated vertices in the remainder. These large components are unique and the three cases are disjoint. Every other graph has largest component at most $n-3$. For $n \geq 7$, its components can be partitioned into two groups of sizes between 3 and $n-3$: use a component of size at least three if present; otherwise greedily accumulate components of size at most two until reaching size three or four. Hence the remaining count is at most

$$\sum_{k=3}^{n-3} I_k I_{n-k} \leq \sum_{k=3}^{n-3} F_k F_{n-k} = O(F_{n-3}),$$

by Lemma 3.1. We have proved the sharper component identity with bounded remainder

$$I_n = C_n + C_{n-1} + 2C_{n-2} + O(F_{n-3}). \quad (43)$$

The connected formula implies

$$C_{n-1} = \tfrac{1}{2}F_{n-1} - F_{n-2} + o(F_{n-2}), \quad 2C_{n-2} = F_{n-2} + o(F_{n-2}).$$

Their F_{n-2} terms cancel, giving (33) and completing the proof of Theorem 9.1.

Corollary 9.2. *With b_0 as in (31),*

$$\frac{2C_n}{F_n} = 1 - \frac{2}{qn} + \frac{q^{-1} - 2q^{-2}(\log n + b_0)}{n^2} + o(n^{-2}), \quad (44)$$

$$\frac{2I_n}{F_n} = 1 - \frac{1}{qn} + \frac{(2q)^{-1} - 2q^{-2}(\log n + b_0)}{n^2} + o(n^{-2}). \quad (45)$$

Proof. The known first Fishburn correction suffices: dividing (7) at consecutive indices gives

$$\frac{F_{n-1}}{F_n} = \frac{1}{qn} \left(1 - \frac{1}{2n} + O(n^{-2}) \right).$$

Indeed $\sqrt{1 - 1/n} = 1 - 1/(2n) + O(n^{-2})$, and the quotient of the factors $1 + a_F/(n - 1) + O(n^{-2})$ and $1 + a_F/n + O(n^{-2})$ is $1 + O(n^{-2})$. Combine this with the second ratio in (15) and Theorem 9.1. The induced logarithmic ratio error is $O(\log n/n^3) = o(n^{-2})$. This determines the nonlogarithmic coefficients in the exact $F_n/2$ normalization without needing a second Fishburn coefficient. $\square$

10 Inverting a counting threshold

For either sequence $a_n = I_n$ or $a_n = C_n$, define

$$N_a(T) = \min\{n \geq 1 : a_n \geq T\} \quad (T > 1).$$

Use the corresponding α from Corollary 1.2. Define, for real $y > 0$,

$$\Phi_\alpha(y) = \log K + \log \Gamma(y + 1) + \frac{1}{2} \log y + y \log q + \frac{\alpha}{y} - \beta \frac{\log y}{y^2}. \quad (46)$$

For all sufficiently large T , there is a unique large solution $y = y_\alpha(T)$ of $\Phi_\alpha(y) = \log T$. The phrase “large solution” means the one in an eventual interval on which Φ_α is strictly increasing; no claim is made about small positive solutions.

Theorem 10.1. *There are constants $B, T_0 > 0$, depending only on the selected sequence, such that for $T \geq T_0$,*

$$\left\lceil y - \frac{B}{y^2 \log y} \right\rceil \leq N_a(T) \leq \left\lceil y + \frac{B}{y^2 \log y} \right\rceil. \quad (47)$$

Let $L = \log T$, let W be the positive real Lambert function, and set

$$x = \frac{L}{W(qL/e)}. \quad (48)$$

Starting with $x_0 = x$, take two Newton steps

$$x_{j+1} = x_j - \frac{\Phi_\alpha(x_j) - L}{\Phi'_\alpha(x_j)}, \quad j = 0, 1. \quad (49)$$

Then

$$x_2 - y = O\left(\frac{1}{x^3(\log x)^3}\right). \quad (50)$$

One may therefore replace y by x_2 in (47), with width $B'/(x^2 \log x)$ for a suitable constant B' .

Proof. First the counts are nondecreasing. Adding an isolated vertex injects interval-graph types of size n into those of size $n + 1$: all isolated vertices are interchangeable, so removing any one recovers the original type. Adding a universal vertex preserves interval graphs, represented by an interval covering all existing intervals, and yields a connected graph. All universal vertices are interchangeable, so this operation injects all interval-graph types of size n into connected types of size $n + 1$. Hence

$$C_{n+1} \geq I_n \geq C_n. \quad (51)$$

In particular both thresholds are well-defined, since (6) tends to infinity.

Taking logarithms in (6) gives

$$\log a_n = \Phi_\alpha(n) + O(n^{-2}). \quad (52)$$

The square of the first correction is absorbed in this error, so its coefficient need not be known. Standard Stirling and digamma estimates, or termwise differentiation of the classical smooth Stirling expansion for $\log \Gamma$, give

$$\Phi_\alpha(t) = t \log(qt/e) + \log t + b + \frac{\alpha + 1/12}{t} - \beta \frac{\log t}{t^2} + O(t^{-3}), \quad (53)$$

$$\Phi'_\alpha(t) = \log(qt) + O(t^{-1}), \quad \Phi''_\alpha(t) = t^{-1} + O(t^{-2}), \quad (54)$$

where $b = \log(K\sqrt{2\pi})$. In particular Φ_α is eventually increasing and unbounded, proving existence and uniqueness of its large root.

Choose M so that the error in (52) is at most M/n^2 beyond an index n_0 . The smooth functions $\Phi_\alpha(t) \pm M/t^2$ are also eventually increasing. Their large roots at level L are, respectively, r_{low} for the plus sign and r_{high} for the minus sign. By the mean value theorem and (54),

$$r_{\text{low}} = y + O\left(\frac{1}{y^2 \log y}\right), \quad r_{\text{high}} = y + O\left(\frac{1}{y^2 \log y}\right), \quad r_{\text{low}} \leq y \leq r_{\text{high}}.$$

For sufficiently large T , all earlier finite counts are smaller than T . Every integer below $\lceil r_{\text{low}} \rceil$ then has $a_n < T$, while the integer $\lceil r_{\text{high}} \rceil$ has $a_n \geq T$. Increasing a common constant in these root estimates proves (47). This argument keeps the rounding inside the brackets, including threshold equalities.

The Lambert definition satisfies $x \log(qx/e) = L$. Equation (53) gives

$$\Phi_\alpha(x) - L = \log x + b + O(x^{-1}).$$

Its derivative is comparable to $\log x$ on any fixed-width neighborhood of x . Comparing Φ_α at $x \pm D$ for a sufficiently large fixed D shows $y = x + O(1)$. In this neighborhood the second derivative is $O(x^{-1})$ and the first derivative is bounded below by a positive constant times $\log x$. Taylor's theorem around x_j , using $\Phi_\alpha(y) = L$, gives the explicit Newton error inequality

$$|x_{j+1} - y| \leq \frac{A}{x \log x} |x_j - y|^2 \quad (55)$$

for a fixed A and all sufficiently large x . The first step has error $O((x \log x)^{-1})$ and the second has error $O(x^{-3}(\log x)^{-3})$. Both remain in the fixed-width neighborhood, justifying the estimates used. This error is $o(x^{-2}/\log x)$, smaller than the asymptotic inversion uncertainty. Since $y/x \rightarrow 1$, it is absorbed in the final bracket width. $\square$

For actual computation the derivative in (49) is

$$\Phi'_\alpha(y) = \psi(y+1) + \frac{1}{2y} + \log q - \frac{\alpha}{y^2} + \beta \frac{2 \log y - 1}{y^3}, \quad (56)$$

where ψ is the digamma function. A simpler one-step approximation, at lower precision, follows from (53):

$$v = x - \frac{\log x + b}{\log(qx)}, \quad y = v + O((x \log x)^{-1}).$$

The two-step version retains the proved logarithmic correction.

Remark 10.2 (Rounding and effectivity). The theorem asserts the existence of constants and an onset; the report does not supply a certified numerical value for either. It is therefore not a finite algorithm that certifies an exact threshold for every given T . Away from the shrinking neighborhoods of integers the two ceilings agree, but near an integer exact counts or effective error bounds are needed. In particular an expression such as $N_a(T) = \lceil y \rceil + O(y^{-2}/\log y)$ would discard the relevant rounding ambiguity and is not asserted.

Remark 10.3 ([write] Monotonicity, and the inverse in the terms of the transseries volume, 5 October 2026). (a) *Strict monotonicity.* $I_{n+1} > I_n$ for $n \geq 1$, and $C_{n+1} > C_n$ for $n \geq 2$ (while $C_1 = C_2 = 1$). Indeed, for $n \geq 1$ adding an isolated vertex gives a disconnected graph, so the injection used for (51) misses the connected interval graph K_{n+1} , and $I_{n+1} \geq I_n + 1$. For $n \geq 2$ the edgeless graph on n vertices is a disconnected interval graph, so $I_n \geq C_n + 1$, and $C_{n+1} \geq I_n \geq C_n + 1$ by (51). Hence $(I_n)_{n \geq 1}$ and $(C_n)_{n \geq 2}$ are strictly increasing and unbounded, and for $T > 1$ the threshold $N_a(T)$ is the integer staircase N_* of `p0:def:three-inverses` in the transseries volume [8], with $n_1 = 1$ for I and $n_1 = 2$ for C (as $C_1 = 1 < T$, the minimum over $n \geq 1$ equals that over $n \geq 2$). Admissible interpolations exist (piecewise linear, for example), and `p0:thm:staircase(1)` gives $N_a(T) = \lceil \nu(T) \rceil$ for each of them.

(b) *The Lambert start is an instance.* The point x of (48) solves $x \log(qx/e) = L$, the dominant block of (53). This is the equation $X(\kappa \log X + d) = L$ of `p0:prop:factorial-core` with

$$\kappa = 1, \quad d = \log q - 1 = \log \frac{6}{e\pi^2}.$$

That proposition gives $v = W_0(Le^d/\kappa) = W_0(qL/e)$ and $X = e^{v-d} = e^v e/q$; as $ve^v = qL/e$, $e^v = qL/(ev)$ and $X = L/v = x$. It is the unique solution with $\log X + d > 0$; here $\log x + d = L/x > 0$.

(c) *The rest is analogous, not an instance.* The equation $\Phi_\alpha(y) = L$ itself, with its $\log \Gamma$, $\frac{1}{2} \log y$, α/y and $\beta \log y/y^2$ terms, is not one of the volume's core equations, and the two Newton steps are a numerical refinement controlled by (55), not the volume's formal reversion (`p0:thm:lambert-centered`). The bracket (47) is proved directly, by comparing a_n at the integers with the two smooth functions $\Phi_\alpha \pm M/t^2$; it uses no interpolation of (a_n) and is not an instance of `p0:thm:staircase`, whose separation condition (2) is the corresponding statement for an interpolated inverse. No novelty is claimed for the inversion.

11 Finite verification and reproducibility

The accompanying standard-library Python program generates Fishburn matrices directly from their nonnegative triangular entries, requiring nonzero rows and columns. It constructs the associated orders and canonicalizes their comparability graph fibers. The graph canonicalization refines degree

classes by neighbor counts and minimizes over the remaining permutations, identifying only true twin permutations as redundant. Poset canonicalization uses the sizes of nested strict down-sets and up-sets; equal pairs identify interchangeable cell elements.

The following values are generated rather than read from a sequence file. The displayed T_n and R_n use diagonal weight and occupancy, respectively.

n	F_n	I_n	C_n	O_n	T_n	R_n
1	1	1	1	1	1	1
2	2	2	1	1	4	3
3	5	4	2	2	14	10
4	15	10	5	6	51	37
5	53	27	15	23	204	153
6	217	92	56	104	911	707
7	1014	369	250	534	4546	3635
8	5335	1807	1328	3051	25212	20666
9	31240	10344	8069	19155	154267	129055

An independent brute-force enumeration of order automorphisms verifies that the occupied cells are exactly the vertex orbits through size five. Through size nine the program checks the simplicial/diagonal characterization, the identity $T_n - R_n = T_{n-1}$, order and graph counts, self-duality, and the complete graph fibers. Through quotient size seven it checks genericity, the singleton-root condition, and the dominant inflation construction:

Quotient m	Graph size n	$ \mathcal{E}_m $	$ \mathcal{D}_n $	Order types per fiber
5	7	8	4	4
6	8	52	26	4
7	9	380	190	4

The second exact family in Section 9 is also checked exhaustively: for core sizes $m = 5, 6, 7$ there are respectively 6, 30, 188 generic non-self-dual cores, mapping two-to-one onto 3, 15, 94 serial-family graphs of sizes 7, 8, 9. Each full serial fiber has four order types, two dual orbits, no self-dual type, a unique universal vertex, and a unique leaf. The two exact families are disjoint. All inside-module, outside-module, and autonomous-pair contractions used in the sharper rooted bounds are checked for retained simpliciality through size seven.

Each displayed dominant graph has exactly two dual orbits, no self-dual order type, and exactly two eligible rooted quotient preimages. Every positive-loss connected fiber through size nine is also classified into the four remainder cases of Section 7 or the dominant family. The program checks every order type within these fibers, including universal-vertex removal and restoration.

Additional checks reconstruct F_n from (23) and I_n from (30), verify the rational-polynomial normalization of the first and logarithmic coefficients, the exact- $F_n/2$ constant conversion, and the ordinal and graph-component contributions at the next shift, and test the Newton derivative, local Taylor remainder, and two-step inversion at finite real inputs. These are consistency checks on the proof and its implementation. They do not establish the infinite asymptotic estimates, a finite error constant, or a rounding certificate.

The source archive contains this article, its PDF, the checker, deterministic build and packing commands, an integrity verifier, and adversarial tests. The verifier uses an exact flat inventory

and hashes every file except its own manifest seal. The manifest is the declared trust anchor, so accidental or adversarial changes require a trusted external archive or manifest digest to detect coordinated resealing. Every check remains active under Python’s optimized mode. Commands refuse existing output targets, in-bundle output destinations, and symlinked path components; reader commands write only to new external locations. The explicit author-only sealing command, which exclusively creates the in-bundle manifest, is the documented exception. The README records the exact commands and build environment. The archive contains no third-party source PDFs; the bibliography provides public links.

[write] **The companion here, and the intake reruns, 5 October 2026.** “The source archive” is the delivered package. In this report `report133.tex` is `article.tex`; the six programs `check_math.py`, `verify.py`, `output_guard.py`, `build.py`, `pack.py` and `test_adversarial.py` are in `code/`; `build-environment.txt` is in `data/`; the delivered README was replaced by a report README; the PDF and the manifest are not shipped. The verifier enforces the delivered flat eleven-file inventory, so it rejects the shipped layout and does not run in the shipped directory; the report README gives two routes, from the arrival commit’s archive and from copies of the shipped programs. At placement, on copies, the checker passed to standard output in normal mode (2 min 26 s) and with `-O` (1 min 19 s), with byte-identical output (13,790 bytes), and `verify.py -inventory-only` passed (10 hashed files). At the write, on a fresh extraction of the archive, `-inventory-only` passed again, and `check_math.py` run with `-O` from copies of the two shipped files `code/check_math.py` and `code/output_guard.py` reproduced that output byte for byte (20 s). The adversarial tests, the PDF build and the packing were not run: they need POSIX-only file calls and the recorded Debian pdfTeX.

12 Source boundaries and further questions

Hwang and Jin [1] supply the Fishburn expansion and the two statistical inputs used here. Gallai’s decomposition, in the presentation [2], supplies the orientation structure. Their use does not transfer an interval-order asymptotic to graphs automatically: the root statistics and graph-fiber losses require the arguments in Sections 4–8.

Bukh and Jeffs [3], in the unlabeled-case remark on printed page 11 of the cited version, obtain the logarithmic estimate

$$\log I_n = n \log n - \left(1 + \log(\pi^2/6)\right) n + O(\log n).$$

The leading equivalent in this report is consistent with, and sharper than, that stated bound. Their nearby principal theorem concerns labeled graphs, a different enumeration problem. The remark itself does not supply the half-Fishburn equivalent or the correction coefficients proved here.

Remark 12.1 ([write] What Bukh and Jeffs state, and the explicit $O(\log n)$ term, 5 October 2026). In [3] (arXiv version 2, 28 June 2023) the unlabelled case is the third item of Section 5, “Problems and remarks”, on p. 11. The proof of their Sandwich Theorem (Theorem 9, for labelled graphs) yields an injection from unlabelled interval orders on $n - 2$ elements to doubly rooted interval graphs on n vertices, whence $F_{n-2}/(n(n-1)) \leq I_n \leq F_n$ in the notation here; the factor $1/(n(n-1))$ “can be upgraded” to $\frac{1}{2}$ by appealing to Hanlon’s work, and the enumeration of unlabelled interval orders in their reference [4] (Brightwell and Keller, Theorem 1) gives the display above. The source quotes that display correctly, and its sentence is accurate: the leading equivalent is consistent with that estimate and sharper. Precisely, for $a_n \in \{I_n, C_n\}$,

$$\log a_n = n \log n - \left(1 + \log(\pi^2/6)\right) n + \log n + \log(K\sqrt{2\pi}) + O(n^{-1}),$$

with $\log(K\sqrt{2\pi}) = 1.2206464656\dots$; so the $O(\log n)$ term of [3] is $\log n + \log(K\sqrt{2\pi}) + O(n^{-1})$, and the same holds for the connected count, which [3] does not treat.

Proof. By (6), $a_n = Kn!\sqrt{n}q^n(1 + O(n^{-1}))$, so $\log a_n = \log K + \log n! + \frac{1}{2}\log n + n\log q + O(n^{-1})$. Stirling's formula gives $\log n! = n\log n - n + \frac{1}{2}\log(2\pi n) + O(n^{-1})$, and $\log q = -\log(\pi^2/6)$. Adding, the two halves of $\log n$ combine into $\log n$ and the constants into $\log K + \frac{1}{2}\log(2\pi)$. $\square$

Observation, not a claim. With the exact I_n , the difference between $\log I_n$ and the right-hand side without its $O(n^{-1})$ is $-0.450, -0.178, -0.102$ at $n = 10, 20, 30$. The elementary bounds $(F_m - B_m^{\text{all}})/2 \leq C_m \leq I_m \leq (F_m + S_m)/2$ of Section 5, with (18), (9) and (7), already give the display above: they show $a_n = (F_n/2)(1 + O(n^{-1}))$.

Hanlon [4] is relevant to exact generating-function enumeration. The full primary article was not available for inspection in preparing this report; no claim here relies on an unverified theorem or terminology from it. The limited source comparison, including the cited OEIS entries, is not an exhaustive novelty search and does not certify historical priority.

Remark 12.2 ([write] Hanlon (1982) and the literature to 2023, as the write read it, 5 October 2026). The write read Hanlon's article [4] (the AMS scan, 44 pages). It enumerates unlabelled, labelled, identity and transitive interval graphs and several kinds of unit interval graphs through a structural decomposition (reduced graphs and their “buried” subgraphs; Theorem 1 shows that a “dependable” interval graph has exactly one interval representation up to the appropriate equivalence), gives the generating functions of I_n and C_n recursively (Theorem 3: the component identity (30) and a substitution equation), and tabulates I_n for every $n \leq 30$ and C_n at selected $n \leq 30$ (Section 9, Tables VII and VIII). It derives asymptotics only for unit interval graphs (Section 7, Theorem 12). For interval graphs its concluding Section 8 (p. 420) lists the asymptotic number as an open problem: Cohen, Komlós and Mueller had probabilistic information but no asymptotic expansion, and it was “not even known whether $I(y)$ has a positive radius of convergence”. Yang and Pippenger [7] (abstract read) answered that question: the ordinary generating function has radius of convergence zero. Acan [6] proved $\log I_n \sim n \log n$ and wrote in his Remark 2 that finding I_n asymptotically would be a strong result; Bukh and Jeffs determined $\log I_n$ up to $O(\log n)$ (Remark 12.1). None of these four sources states the equivalent $I_n \sim F_n/2$, a statement for C_n , or a correction term. This settles the source's caveat about Hanlon: the leading term is not in Hanlon's article, which poses the question. It does not establish priority: the literature after mid-2023 was not searched exhaustively, and the write did not compare Hanlon's decomposition in detail with the modular decomposition used here. The rows of Hanlon's Table VII that the scan renders legibly agree with the OEIS terms of A005975, and its six further rows ($n = 25, \dots, 30$), inverted through (30), reproduce the connected counts of Table VIII at $n = 10, 15, 20, 25, 30$ exactly; they are used in Question 1. (Added after the independent check of 5 October 2026: the OEIS b-files of A005975 and A005976 already list $n \leq 30$; Table VII's rows 25, $\dots$, 30 equal the A005975 b-file, and the inversion of that b-file reproduces the A005976 b-file at every $n \leq 30$.)

The exact shifted formulas determine b_0 . The nonlogarithmic coefficient under the conventional $Kn!\sqrt{n}q^n$ normalization is not evaluated here: it also involves the second coefficient in the Fishburn expansion. The inverse theorem retains its stated, valid error scale. Several questions remain within this report's scope.

1. Quantify the remainder beyond the now identified shifted-Fishburn constant. This requires sharper control of excluded rooted families and small-module interactions, rather than the diagonal mean alone.
2. Determine whether a full power-logarithm expansion exists. The present proof establishes the

logarithmic term and its following shifted constant, and gives no all-orders theorem.

3. Obtain effective constants and an explicit onset in the remainder. These would turn the inverse brackets into certified finite threshold computations.
4. Analyze higher exceptional graph fibers and their interactions. A systematic description may explain later logarithmic powers, but it must keep rooted isomorphism classes distinct from vertex multiplicities.

The finite computations are deliberately separated from these questions: agreement at small sizes verifies exact combinatorial constructions, not asymptotic priority or an unproved extension.

12.1 Further questions and research

[write] **Added 5 October 2026, batch 106.** Following Vladimir Reshetnikov’s standing rule of 4 October 2026, every claim of the source that is not proved in full is stated here as an open question, with its source, the argument offered and what is missing. The four items above are Questions 1–4; Question 5 is the unevaluated coefficient named in the paragraph above them, and Question 6 concerns the external inputs and priority. Neither the intake nor the write found a false claim in the source.

1. **The remainder beyond b_0** (source, item 1). Theorem 9.1 gives $o(F_{n-2})$ with no rate. Missing: sharper control of the excluded rooted families and of small-module interactions. In the computation below the rooted diagonal mean alone approaches its limit only like $(\log m)/m$.

Observations, not claims. Put $\delta_n^C = (F_n/2 - F_{n-1} - C_n)/F_{n-2} - \log n$ and $\delta_n^I = (F_n/2 - F_{n-1}/2 - I_n)/F_{n-2} - \log n$; by Theorem 9.1 both tend to $b_0 = 1.0795154\dots$. From the OEIS terms ($n \leq 24$ for I , $n \leq 25$ for C) and Hanlon’s Table VII for $n = 25, \dots, 30$ (Remark 12.2; the OEIS b-files of A005975 and A005976, fetched 5 October 2026, list $n \leq 30$ and agree with Table VII and with its inversion; this parenthesis was added after the independent check of 5 October 2026):

n	15	20	24	25	26	27	28	29	30
δ_n^C	1.2755	1.1911	1.1053	1.0914	1.0800	1.0707	1.0631	1.0570	1.0520
δ_n^I	1.8802	1.6329	1.4642	1.4348	1.4095	1.3876	1.3685	1.3518	1.3371

δ_n^C decreases from its maximum 1.2817 at $n = 16$ through $n = 30$ and falls below b_0 between $n = 26$ and $n = 27$; δ_n^I decreases from $n = 15$ and is still above b_0 at $n = 30$. The intake, which had the values through $n = 25$, read them as decreasing towards b_0 ; the five further terms show that δ_n^C does not approach b_0 monotonically from above. Fits of $b + (c_1 \log n + c_0)/n$, with or without $(\log n)^2/n$ and n^{-2} terms, each through as many consecutive values of δ_n^C or δ_n^I (three to five) as it has parameters, ending at $n = 26, 28$ or 30 , give limits between -21.4 and 2.8 (between -3.8 and 1.7 for the windows ending at $n = 30$) that move with the window. (Corrected after the independent check of 5 October 2026; the first wording read “through the last three to five values give limits between -21.4 and 2.8 ”, but those extremes come from windows ending at $n = 26$.) Thirty terms neither confirm nor contradict b_0 , and a crossing does not contradict Theorem 9.1, whose remainder may change sign. One input can be tested much further: with T_m computed exactly through $m = 200$ (inclusion–exclusion over forced zero rows and columns, which reproduces the T_n of Section 11), $T_m/F_m - 2 \log m = 0.338$,

0.283, 0.201 at $m = 30, 50, 200$, and fits of $b + (c_1 \log m + c_0)/m + c_2(\log m)/m^2$ give 0.15904, against $2a_0 = 2(\log q + \gamma) = 0.159031$ in (34). This tests that input only; at $m = 30$ it is still 0.18 from its limit, so the remainders are large at the sizes of the table. (Added after the independent check of 5 October 2026: that check computed T_m exactly through $m = 200$ by a second route, the bivariate generating function of the diagonal in Proposition 4(ii) and (2.12) of [1], differentiated at $v = 1$, and found the same values; with an additional m^{-2} term the fits on five points ending at $m = 160$ and at $m = 200$ give 0.159031, that is $2a_0$ to about $2 \cdot 10^{-7}$.)

2. **A full power–logarithm expansion** (source, item 2). The proof gives the logarithmic term and the following shifted constant only. Missing: every further term, which would involve the later Fishburn coefficients and the exceptional fibres of Question 4.
3. **Effective constants and an explicit onset** (source, item 3, and the remark after Theorem 10.1). With them the brackets (47) would become certified finite computations; without them $N_a(T)$ near an integer needs exact counts, which are known through $n = 30$. Missing: an explicit error constant and onset in (7) and (8) and in every O of Sections 3–8.
4. **Higher exceptional fibres** (source, item 4). A systematic description of graph fibres with more than one dual orbit, keeping rooted isomorphism classes distinct from vertex multiplicities, may explain later logarithmic powers. Missing: the classification beyond the two exact families $\mathcal{D}_n$ and $\mathcal{J}_n$.
5. **The n^{-2} coefficient in the conventional normalization** (source, paragraph above the list). In (6) the coefficient of n^{-2} is not evaluated; Corollary 9.2 gives the nonlogarithmic coefficients in the $F_n/2$ normalization. Missing: the second coefficient of the Fishburn expansion, which (1.1) of [1] as cited does not print, and the bookkeeping that combines it with Corollary 9.2.
6. **External inputs and priority.** The proofs use equation (1.1), Theorem 25(ii) (with the probability generating function uniform for real v near 1) and Corollary 29 of [1], and Gallai’s theorems as presented in [2]; the write located them and did not re-derive them. Hanlon’s article does not contain the leading equivalent (Remark 12.2), but priority beyond the sources read is open; the derivation of $I_n \sim F_n/2$ from Gallai’s theorems and the rarity of nongeneric orders (Section 5) is short, and Bukh and Jeffs’s remark that their factor can be upgraded to $\frac{1}{2}$ through Hanlon’s work points the same way.

[write] **Independent check of the write, 5 October 2026.** An adversarial check made by the intake after the write (commit c6488293e) fetched the OEIS b-files of A005975, A005976 (both $n = 1, \dots, 30$) and A022493 ($n \leq 488$) and reread Hanlon’s article, the Bukh–Jeffs and Acan preprints and the Yang–Pippenger abstract. It confirmed Remark 12.2 against those texts and the assignment of the two sequences (A005976 counts the connected graphs). The inverse Euler transform of the A005975 b-file reproduces the A005976 b-file at every $n \leq 30$, and the inverted values equal Hanlon’s Table VIII at every printed row; Table VII’s rows 25, $\dots$, 30 equal the A005975 b-file. It confirmed Remark 12.1 (the reading of Bukh and Jeffs’s printed p. 11, the derivation from Corollary 1.2 and the differences $-0.450, -0.178, -0.102$) and the three parts of Remark 10.3, with the monotonicity checked on the data through $n = 30$. All eighteen values of δ_n^C and δ_n^I in Question 1 agree, as do the maximum at $n = 16$ and the crossing between $n = 26$ and $n = 27$. Two sentences of that question were corrected, with dated notes keeping the first wording; the values beyond the OEIS data fields are also in the OEIS b-files, and the fit range -21.4 to 2.8 comes from windows ending at $n = 26$ (the windows ending at $n = 30$ give -3.8 to 1.7); the conclusion is unchanged. It computed T_m through $m = 200$ independently, from the diagonal generating function of Hwang and Jin (checked against brute force for $m \leq 6$ and with F_m equal to the A022493 b-file for $m \leq 200$), confirmed 0.338, 0.283, 0.201 and 0.15904, and with an additional m^{-2} term obtained $0.159031 = 2a_0$. No mathematical error was found.

References

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- [3] B. Bukh and R. Jeffs, *Enumeration of interval graphs and d -representable complexes*. arXiv:2203.12063v2; unlabeled-case remark, printed page 11. arXiv record.
- [4] P. Hanlon, *Counting interval graphs*, Transactions of the American Mathematical Society 272 (1982), 383–426. DOI record. Included for context; full primary text not used as a proof input.
- [5] OEIS Foundation Inc., A005975, A005976, and A022493. Sequence identifiers and comparison data, consulted 2 October 2026.
- [6] H. Acan, *Counting unlabeled interval graphs*. arXiv:1810.02040v1 (2018). arXiv record. [Added by the write; read in full.]
- [7] J. C. Yang and N. Pippenger, *On the enumeration of interval graphs*. Proc. Amer. Math. Soc. Ser. B 4 (2017), 1–3. [Added by the write; abstract only.]
- [8] ProveIt, *Transseries and inversion* (the transseries volume), Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex: p0:prop:factorial-core, p0:def:three-inverses, p0:thm:staircase, p0:thm:lambert-centered. [Added by the write.]